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ASSIGNMENT 3: INTELLIGENT AGENT

THEME: SMART EDUCATION

LECTURER: DR. ROLIANA BINTI IBRAHIM

NAME	**MATRIC NO**
YASMIN BATRISYIA BINTI ZAHIRUDDIN	A23CS0201
NABIL AFLAH BOO BINTI MOHD YOSUF BOO YONG CHONG	A23CS0252
SYARIFAH DANIA BINTI SYED ABU BAKAR	A23CS0183
AIN NURNABILA BINTI MOHD AZHAR	A23CS0207

TABLE OF CONTENTS

1.0 PEAS Model Representation

Performance: How well students are learning, their engagement levels, improvement in scores, how quickly the system catches struggling students, accuracy in detecting weak areas, and how fast lecturers can respond to alerts

Environment: The Learning Management System (LMS), student databases, assessment records, attendance systems, classroom analytics, and past performance data

Actuators: Assigning remedial modules, delivering practice exercises, moving students to next topics, triggering motivational breaks, sending alerts to lecturers, and monitoring engagement

Sensors: Student data input, proficiency score checker, engagement detector (tracks attendance, participation, LMS activity), assessment history analyzer, and prerequisite checker

1.1 Explanation on each property in PEAS model

Performance Measure, Think of this as your system's report card. How do you know if it's doing a good job? You look at whether students are actually improving, staying engaged with their studies, and if lecturers are getting warned early enough when students start struggling. If the system can catch a student falling behind before it's too late, that's a win.

Environment is where the system "lives" and works. It's hanging out in the LMS, reading through grade databases, checking attendance records, and looking at how students have performed over time. The environment keeps changing because students' performance and engagement fluctuate throughout the semester.

Actuators are the system's hands and what it can actually do. When it spots a struggling student, it can assign them extra help through remedial modules. If they're doing okay, it gives them practice exercises. For the advanced students, it moves them to harder material. If someone's zoning out, it throws in a motivational break. And when things look serious, it rings the alarm bell to notify the lecturer.

Sensors are the system's eyes and ears and how it gathers information. It reads student data, analyzes test scores, watches engagement patterns, for instance, like who's showing up to class and participating, checks how students have been doing over time, and verifies if they have the right background knowledge. All this info helps the system decide what to do next.

2.0 Relations between each property

This section will show the relations between each property in the PEAS model including performance measure, environment, actuators, and sensors.

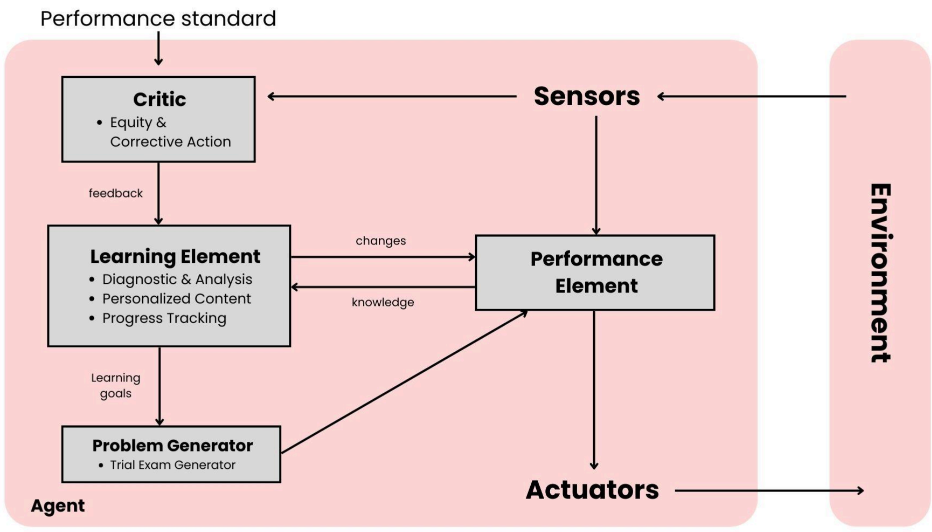

Figure 2.0 Intelligent Agent of Smart Education

The Figure 2.0 utilizes a Learning Agent Architecture which enables a continuous feedback loop between student data and system adaptation. The learning element serves as the system's brain by integrating the Diagnostic and Analysis, Personalized Content, and Progress Tracking agents. Sensors like marks of quizzes and student engagement levels will be analyzed to identify the knowledge gaps and provide the learning path that is suitable for the students. This helps to ensure the system meets its primary goals of early gap identification and maintain a flexible and personalized curriculum.

The performance element will execute the task such as suggest suitable modules and notifying the lecturers through the system's actuators. While the performance element handles the immediate execution of the student learning journey, the critic that is represented by the Equity and Corrective Action agent will help to monitor the long-term success. The critic will compare the student progress against the performance standard by detecting its pattern and send feedback to the learning element to trigger the corrective actions and additional support.

The problem generator consists of a Trial Exam Generator to challenge the student with the new assessments to test their "final exam readiness". The data that have been gathered from these trial examinations will feed back to the sensors which allows the system to refine its diagnostic accuracy.

In all, these components help to ensure the agent actively learns and improves to reduce performance variance across the student population.

3.0 Explanation in Proof of Concept (POC)

Proof of Concept (POC) demonstrate the feasibility of the suggested Smart Education Intelligent Agent by demonstrating how the system functions in a controlled academic environment utilizing actual student data inputs and simulated learning scenarios. The POC's goal is to confirm that the agent can successfully detect learning gaps early on, customize learning paths, let instructors track students' progress and lessen performance gaps.

Historical academic records, test results, attendance records, and engagement data are the main inputs used in the POC's implementation within a Learning Management System (LMS) context. These data points are continuously gathered by sensors and fed into the Diagnostic & Analysis Agent. In addition to determining the goal scores necessary for students to receive an A grade, this agent evaluates assessment trends and prerequisite mastery to categorize each topic as strong, average, or weak.

The Personalized Content Agent is triggered if knowledge gaps are identified. The bot suggests appropriate remedial modules, focused practice tasks, or advanced learning resources based on prerequisite mapping and sub-topic performance. Instead of a curriculum that is one-size-fits-all, this guarantees that every student obtains a personalized learning path. The main performance metric used to confirm the efficacy of this customisation is an improvement in post-module assessment scores.

The Trial Exam Generator creates adaptive trial exams with past year questions and difficulty settings that are in line with students' areas of weakness in order to assess student preparedness and system correctness. The results are returned to the sensors for additional processing, and instant scoring enables prompt feedback. The dependability of the system's diagnostic skills is confirmed by looking at the relationship between trial exam performance and anticipated final exam preparation.

Progress Tracking Agent regularly monitors student involvement, including attendance, time spent on learning modules and participation in examinations,. Through the communication API, the agent notifies professors when abnormal patterns are identified, such as inactivity or persistent poor performance. This enables instructors to step in early, offer advice or provide more assistance before pupils drastically lag behind.

The Equity & Corrective Action Agent serves as the system's critic to guarantee equity and long-term progress. It finds growing differences between high-performing and struggling children by comparing student growth to predetermined performance standards. The agent autonomously modifies learning paths when performance variance is identified, giving weaker students more resources, more practice, or different explanations. The learning aspect receives feedback from this agent, enabling the system to continuously improve its tactics.

Overall, the POC verifies that the intelligent agent is capable of functioning as a closed-loop learning system. The system responds in real time to the needs of students through ongoing sensing, analysis, action, and feedback. The POC's findings show that the suggested design may accomplish the system's objectives of equitable academic achievements, early gap detection, individualized learning, and lecturer support.

4.0 Behaviours of Agent in AI to achieve The System Goals

Table 4.0 shows agents with their PEAS representation.

Agent	Performance measure	Environment	Actuators	Sensors
Diagnostic & Analysis Agent	Precision in identifying knowledge gaps. Accuracy of prediction to get an A	Student academic record. Grading schemes	Classifies topic status. Calculate target marks.	Quiz result. Test marks Historical grade data
Personalized content agent	Improvement in post-module scores	Library of sub-topic. Prerequisite mapping	Suggest specific modules	Prerequisite status. Sub-topic performance
Trial Exam Generator	Correlation between trial result and final exam readiness	Database of Past Year Question	Generate a Trial Exam. Provides instant scoring	Difficulty settings Students weak topic history
Progress Tracking Agent	Accuracy of student Effort reports. Reliability of readiness alerts	Lecturer dashboard. Communication API	Triggers lecturer notification Update students progress logs	Engagement checks Time spent on modules Final trial scores
Equity & Corrective Action Agent	Reduction in performance variance between students	Adaptive learning paths	Routes for struggling students to give an additional support	Repeated weak status inactive / struggle patterns

Table 4.0 agents with PEAS Model

Alignment of PEAS Model with the System Goals :

1. The Diagnostic agent uses its sensors to perform Intelligent Data Analysis. Its Actuator is a specific instruction on what the students are lacking. This aligns with goal 1 which is to Identify gaps early & calculate targets for students.

2. The personalized content agents act as the main job of the state space search where it ensures the learning journey is flexible based on students' needs and this agent is aligned with goal 2 which is to have personalized learning modules.

3. The progress tracking agent uses sensors to monitor students effort and only uses its actuator which gives notifications to the lecturers when students need help for their modules and this agent is aligned with goal 3 which is to assist lecturers in tracking progress.

4. The equity agent ensures that struggling students work harder by automatically providing them materials that they need in order to reach the advanced state. This agent will effectively close the knowledge gap where it aligns with goal 4 which is to reduce the gap between students who have difficulties in understanding and those who are advanced.